# Mohammed Shakib Hossain

*Research Statement*

## Research Overview

My research interests lie at the intersection of usable security and privacy, human-computer interaction (HCI), and human-centered security. I am interested in understanding how people experience and make decisions about security and privacy in the technologies and organizations they encounter in everyday life. In particular, I want to understand why security mechanisms that are technically sound or formally required do not always fit people's needs, behaviors, or practical circumstances.

My research experience has allowed me to explore these questions from different perspectives. My undergraduate thesis focused on authentication in federated metaverse environments and included both technical development and usability evaluation. My later coursework and ongoing master's research have led me to examine how people deal with security requirements in workplace settings. More recently, my HCI research has introduced me to questions about personal information disclosure when people seek advice from AI systems. These experiences have shaped my interest in usable privacy, empirical and behavioral security, and the human factors that influence security decisions. I am also interested in the technical foundations of security, particularly authentication, digital identity, and security protocols.

Through further research, I hope to combine empirical methods with technical analysis and user-centered design. I want to better understand the challenges people face when using security and privacy mechanisms and explore how these mechanisms can be made more suitable for the contexts in which they are used.

## Research Background

My research experience began with my undergraduate thesis, *A Secure Authentication Mechanism for Federated Metaverses Using Non-Fungible Tokens (NFTs)*, conducted at BRAC University under the supervision of Dr. Md Sadek Ferdous. Working with two thesis partners, I contributed to developing the threat model and formally verifying the authentication protocol using ProVerif. We also developed a proof of concept that connected a Next.js and Node.js frontend with MetaMask and a Minecraft Paper server in a VR-based metaverse. In this system, users authenticated through their wallets to gain access to gameplay.

Alongside the technical work, I contributed to a usability evaluation using a Meta Quest 3, the System Usability Scale (SUS), and a custom user-experience questionnaire informed by ISO 9241-11 and Nielsen's heuristics. The evaluation identified difficulties that some participants faced when switching between the VR headset and a desktop wallet. This helped me understand that a security mechanism can work as intended from a technical perspective while still creating difficulties for its users. It also encouraged me to consider usability alongside the technical design of security systems. I led the preparation of the research manuscript, which was accepted for publication at the IEEE International Conference on Blockchain 2026.

My subsequent coursework research expanded my interest from authentication and usability to the

human and organizational aspects of security. For CSE791, Research Methodology and Seminar, I conducted a qualitative study titled *Understanding Security Policy Violations and Exceptions in Organizational Workflow in Bangladesh*. The study involved five semi-structured interviews with software and IT workers, followed by reflexive thematic analysis. It explored how work demands influence employees' choices of communication channels, how security expectations are learned through everyday work, how client requirements affect information sharing, and how organizational responses relate to existing workplace practices. This small exploratory study gave me experience in conducting interviews and analyzing accounts of security-related practices in organizational settings. I also explored related questions through two other coursework projects. In CSE726, Information Assurance and Human Factors, I developed a research project on security policy violations in organizational workflows, with an emphasis on the human and organizational factors that may influence security practices. In CSE773, Usable Security and Privacy, I developed a project titled *Living Under Watch: Privacy, Surveillance, and Digital Harm among Tea Garden Workers in Bangladesh*. This project considered how surveillance, social circumstances, and unequal power relationships may affect people's privacy and experiences of digital harm. Both projects involved developing research questions, reviewing relevant literature, and planning proposed studies; neither produced completed empirical findings. They helped me think about how security and privacy are shaped not only by technology, but also by people's social circumstances and the environments in which they live and work.

## Ongoing Research

My ongoing master's research at BRAC University, supervised by Dr. S M Taiabul Haque, examines how information security policy deviations and exceptions are understood and managed in everyday organizational work in Bangladesh. The research is motivated by the difference that can arise between formal security requirements and the practical demands of workplace activities. I am interested in how human, organizational, and technological factors influence the way employees deal with security requirements, and why certain deviations or workarounds occur in practice.

My preliminary work has included exploratory semi-structured interviews with software and IT professionals in Bangladesh, as well as a review of literature on security policy compliance, workplace security practices, usable security, and human factors. Building on this initial work, I aim to examine how employees navigate security requirements in their daily tasks and how organizational actors understand and respond to deviations and exceptions. The research is ongoing, and its final scope and title are still being refined.

Alongside my master's research, I am working on a coursework research project for CSE731, HCI and Design Research, titled *Before You Share: Designing a User-Controlled Privacy Review Interface for Everyday AI Advice-Seeking Among Bangladeshi University Students*. The project focuses on how university students decide what personal information to include when seeking everyday, non-academic advice from AI systems. Examples include seeking advice about career decisions, interpersonal difficulties, family expectations, and personal planning. I am particularly interested in how students distinguish between information that provides useful context for an advice request and personal details that may reveal more than necessary.

The proposed study will use a mixed-method approach, combining a questionnaire and semi-structured interviews to explore students' disclosure decisions and privacy considerations. Based on the findings, I plan to develop and evaluate a clickable prototype in Figma. The interface will allow users to review individual details in a prompt and decide whether to retain, generalize, or remove them before submitting it to an AI chatbot. The main idea is to help users reduce unnecessary

personal disclosure without removing contextual information they consider important for receiving relevant advice. Importantly, the interface is intended to support users' own decisions rather than decide for them. The project is currently at the proposal and research-design stage; the needs assessment, prototype development, and evaluation are planned.

## Future Research Directions

My research experiences have led me to three related areas that I would like to explore further.

First, I am interested in usable security, privacy, and human-computer interaction, particularly how people understand and use authentication mechanisms and privacy controls. My undergraduate thesis introduced me to the challenges of designing and evaluating authentication systems in immersive environments. Building on this experience, I would like to investigate how security mechanisms can better account for users' needs, expectations, and practical circumstances. I am also interested in these challenges in emerging technologies, including immersive systems and AI-enabled applications.

Second, I want to continue exploring empirical and behavioral security, especially the relationship between formal security requirements and everyday workplace practices. My master's research has motivated me to look beyond policy compliance alone and understand why employees sometimes take alternative approaches to completing their work. I would like to examine how workplace norms, organizational structures, social relationships, and operational demands influence security-related decisions. I am also interested in how organizations can develop security practices that account for the realities of their workflows, particularly in contexts such as Bangladesh and the wider South Asian region.

Third, I am interested in usable privacy and human-AI interaction, particularly how people decide what personal information to share when seeking advice from AI systems. Building on my ongoing HCI project, I would like to study how the purpose of an advice request, the perceived usefulness of contextual information, and privacy concerns influence disclosure decisions. I am especially interested in user-controlled privacy interfaces that help people recognize which details are useful for a particular interaction and which may be unnecessarily revealing. I hope to explore how such designs can support informed decisions while allowing people to remain in control of their personal information.

In terms of methodology, I hope to build on my experience with semi-structured interviews, reflexive thematic analysis, usability evaluation, threat modeling, and formal verification. I also want to develop stronger skills in quantitative and mixed-methods research. Depending on the research question, I am interested in using interviews, surveys, behavioral studies, usability testing, prototyping, and technical analysis. I see these approaches as complementary ways to understand security and privacy challenges from both technical and human perspectives.

In the longer term, I hope to contribute to research that brings together technical analysis and an understanding of people's everyday experiences. I want to investigate security and privacy challenges in real-world settings and contribute to systems and practices that are not only technically sound, but also understandable, usable, and appropriate for the people who rely on them.